NOTE JavaScript defaults to type conversion when using the == test, whereas it forces no type conversion when using === and the like, so you'll find your code will often work fine without casting. But this will come back to haunt you in situations where it doesn't default to the type you expect, for example, when you try to sort an array and JavaScript sorts your numbers alphabetically.

SCALES AND SCALING

Numerical data rarely corresponds directly to the position and size of graphical elements onscreen. You can use `d3.scale()` functions to normalize your data for presentation on a screen (among other things). The first scale we'll look at is `d3.scale().linear()`, which makes a direct relationship between one range of numbers and another. Scales have a domain setting and a range setting that accept arrays, with the domain determining the ramp of values being transformed and the range referring to the ramp to which those values are being transformed. For example, if you take the smallest population figure in cities.csv and the largest population figure, you can create a ramp that scales from the smallest to the largest so that you can display the difference between them easily on a 500-px canvas. In figure 2.5 and the code that follows, you can see that the same linear rate of change from 500,000 to 13,000,000 maps to a linear rate of change from 0 to 500.

You create this ramp by instantiating a new scale object and setting its domain and range values:

```
var newRamp = d3.scale.linear().domain([500000,13000000]).range([0, 500]);
newRamp(1000000);
newRamp(9000000);
newRamp.invert(313);
```

Returns 20, allowing you to place a country with population 10,000,000 at 20 px

Returns 340

The invert function reverses the transformation, in this case returning 8325000.

You can also create a color ramp by referencing CSS color names, RGB colors, or hex colors in the range field. The effect is a linear mapping of a band of colors to the band of values defined in the domain, as shown in figure 2.6.

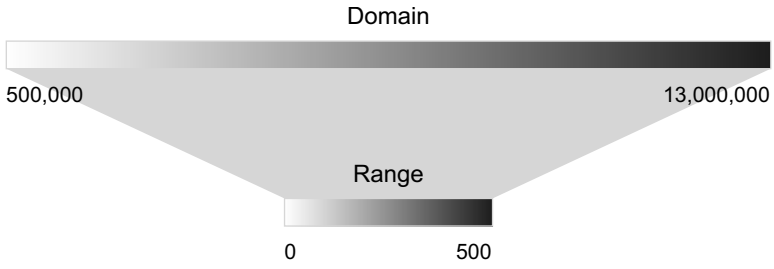


Figure 2.5 Scales in D3 map one set of values (the domain) to another set of values (the range) in a relationship determined by the type of scale you create.